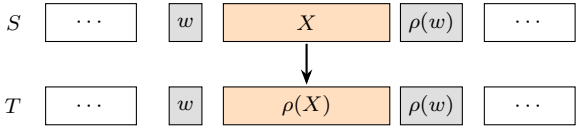
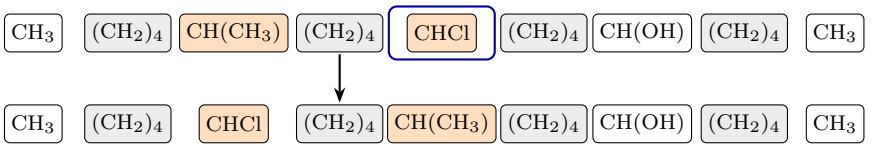
(a)



inversion between the inverted repeat $w \cdots \rho(w)$: $\mathrm{spec}_{k+1}(T)+\mathrm{spec}_{k+1}(\rho T) = \mathrm{spec}_{k+1}(S)+\mathrm{spec}_{k+1}(\rho S)$ for $|w| \geq k$, although a one-strand $(k+1)$-spectrum changes; so $T \in \mathrm{DS}(S, k)$

(b)



exchange of X and Y across identical linkers of length $\ell = 4 \geq 2r - 1$: every radius-2 environment is preserved, so ECFP4 (bits, counts, unhashed) is identical; the topological distance Cl$\cdots$OH is 7 bonds in the upper molecule and 12 in the lower one.